MORPHOLOGY OF FLOWERING PLANTS

NCERT Class 11 Biology — Chapter 5 | High-Yield Deep Examples Master Notes

100% NCERT LINE-BY-LINE | ALL EXAMPLE COLLECTIONS | PLACENTATION & AESTIVATION | FAMILY BREAKDOWN | NEET MCQS

1. THE ROOT (TYPES & MODIFICATIONS WITH EXAMPLES)

Roots develop from the radicle of the embryo (or non-radicle parts in adventitious roots).

Root Type / Modification	Key Structural Feature	NCERT All Examples
Tap Root System	Primary root arises directly from radicle + secondary/tertiary roots. Seen in Dicotyledons.	**Mustard (Sarson)**, Turnip, Gram, Pea
Fibrous Root System	Primary root is short-lived, replaced by a large number of roots arising from base of stem. Monocotyledons.	**Wheat (Gehun)**, Paddy (Rice), Maize
Adventitious Roots	Roots arising from parts of the plant other than the radicle.	**Grass**, **Monstera**, **Banyan Tree**
Storage Roots	Swollen tap or adventitious roots for food storage.	Tap: **Carrot**, **Turnip** Adventitious: **Sweet Potato (Shakarkand)**, Asparagus
Prop Roots	Hanging structures supporting heavy branches.	**Banyan Tree (Bargad)**
Stilt Roots	Supporting roots coming out of the lower nodes of the stem.	**Maize**, **Sugarcane (Ganna)**
Pneumatophores	Vertical upward roots (negatively geotropic) growing in swampy areas for respiration. Have lenticels.	**Rhizophora** (Mangroves)

2. THE STEM (MODIFICATIONS & EXAMPLES)

Stem Modification	Function & Description	NCERT All Examples
Underground Storage	Perennation to tide over unfavorable conditions & food storage.	**Potato (Aloo)**, **Ginger (Adrak)**, **Turmeric (Haldi)**, **Zaminkand**, **Colocasia (Arbi)**
Stem Tendrils	Slender, spirally coiled structures originating from axillary buds for climbing support.	Gourds (**Cucumber**, **Pumpkins**, **Watermelon**), **Grapevines**
Stem Thorns	Woody, straight, pointed structures modified from axillary buds for protection against grazing animals.	**Citrus (Nimbu/Santara)**, **Bougainvillea**
Phylloclade (Photosynthetic)	Fleshy flattened or cylindrical green stem performing photosynthesis in arid regions.	Flattened: **Opuntia** Cylindrical: **Euphorbia**
Sub-aerial Vegetative Spreaders	Runners, Stolons, Offsets, Suckers spreading to new niches.	Runners: **Grass**, **Strawberry** Stolon: **Mint (Pudina)**, **Jasmine** Offset: **Pistia**, **Eichhornia (Water Hyacinth)** Sucker: **Banana**, **Pineapple**, **Chrysanthemum**

3. THE LEAF (PHYLLOTAXY, VENATION & MODIFICATIONS)

🌿 Venation Pattern

- **Reticulate Venation:** Veinlets form a network. Characteristic of **Dicots**.
- **Parallel Venation:** Veins run parallel within lamina. Characteristic of **Monocots**.

🍂 Leaf Modifications

- **Tendrils (Climbing):** **Peas**
- **Spines (Defense):** **Cacti**
- **Fleshy Food Storage:** **Onion**, **Garlic**
- **Phyllode (Photosynthetic Petiole):** **Australian Acacia**

- **Insectivorous Leaves:** Pitcher Plant, Venus Flytrap

Phyllotaxy (Pattern of Leaf Arrangement on Stem):

Phyllotaxy Type	Pattern Description	NCERT All Examples
Alternate	A single leaf arises at each node in an alternate manner.	China Rose (Gurhal), Mustard, Sunflower
Opposite	A pair of leaves arises at each node opposite to one another.	Calotropis (Aak), Guava (Amrood)
Whorled	More than two leaves arise at a node and form a whorl.	Alstonia

4. INFLORESCENCE & FLOWER SYMMETRY

- **Racemose Inflorescence:** Main axis continues to grow continuously; flowers borne laterally in **acropetal succession** (older at base, younger at top).
- **Cymose Inflorescence:** Main axis terminates in a flower (limited growth); flowers borne in **basipetal order** (older at top, younger at base). e.g., Solanum, Jasmine.

Flower Symmetry:

Symmetry Type	Definition	NCERT All Examples
Actinomorphic (Radial)	Can be divided into two equal radial halves in any radial plane passing through the center.	Mustard, Datura, Chilli
Zygomorphic (Bilateral)	Can be divided into two similar halves only in one particular vertical plane.	Pea, Gulmohur, Bean, Cassia
Asymmetric (Irregular)	Cannot be divided into two similar halves by any vertical plane through center.	Canna

Ovary Position relative to Floral Parts:

Ovary Type	Position Description	NCERT All Examples
Hypogynous (Superior Ovary)	Gynoecium occupies highest position, other parts situated below it.	Mustard, China Rose, Brinjal
Perigynous (Half-Inferior Ovary)	Gynoecium situated in center, other parts on the rim of thalamus at same level.	Plum, Rose, Peach
Epigynous (Inferior Ovary)	Margin of thalamus grows upward enclosing ovary completely and fusing with it; other parts above ovary.	Guava, Cucumber, Ray florets of Sunflower

5. AESTIVATION & PLACENTATION (ULTRA HIGH-YIELD)

Aestivation (Arrangement of Sepals / Petals in Floral Bud):

Aestivation Type	Key Structural Feature	NCERT All Examples
Valvate	Sepals/petals touch each other at margin without overlapping.	Calotropis
Twisted	One margin overlaps that of the next one systematically in one direction.	China Rose, Lady's Finger (Bhindi), Cotton
Imbricate	Margins overlap one another but not in any particular direction.	Cassia, Gulmohur
		Pea, Bean (Fabaceae family)

Aestivation Type	Key Structural Feature	NCERT All Examples
Vexillary (Papilionaceous)	5 petals: 1 large standard (posterior), 2 lateral wings, 2 smallest fused keel (anterior).	

Placentation (Arrangement of Ovules inside Ovary):

Placentation Type	Structural Feature	NCERT All Examples
Marginal	Placenta forms a ridge along ventral suture of ovary; ovules borne in two rows.	**Pea**
Axile	Placenta is axial and ovules attached to it in a multilocular ovary.	**China Rose** , **Tomato** , **Lemon**
Parietal	Ovules develop on inner wall of ovary or peripheral part. False septum (**Replum**) formed.	**Mustard** , **Argemone**
Free Central	Ovules borne on central axis and septa are absent.	**Dianthus** , **Primrose**
Basal	Placenta develops at base of ovary and a single ovule is attached to it.	**Sunflower** , **Marigold**

6. PLANT FAMILIES & COMPLETE EXAMPLE COLLECTIONS

Family Name	Key Diagnostic Features & Floral Formula Landmark	Complete NCERT Examples List
Fabaceae (Legume Family)	Zygomorphic, Vexillary aestivation, Diadelphous stamens $(9)+1$, Marginal placentation. $\% \mathbin{\rlap{\bigcirc}\kern-0.1em\raisebox{0.15ex}{\small\textbf{+}}} K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_{1}$	• **Pulses:** Gram, Arhar, Sem, Moong, Soyabean • **Edible Oil:** Soyabean, Groundnut • **Dye:** Indigofera • **Fibers:** Sunhemp • **Fodder:** Sesbania, Trifolium • **Ornamental:** Lupin, Sweet Pea • **Medicine:** Muliathi
Solanaceae (Nightshade / Potato Family)	Actinomorphic, Persistent Calyx (gamosepalous), Epipetalous stamens, Swollen placenta with Axile placentation. $\oplus \mathbin{\rlap{\bigcirc}\kern-0.1em\raisebox{0.15ex}{\small\textbf{+}}} K_{(5)} \overbrace{C_{(5)} A_{5}} \underline{G}_{(2)}$	• **Food:** Potato, Tomato, Brinjal • **Spice:** Chilli • **Medicine:** Belladonna, Ashwagandha • **Fumigatory:** Tobacco • **Ornamental:** Petunia
Liliaceae (Lily / Monocot Family)	Tepals / Perianth $(3+3)$ fused, Epiphyllous / Epitepalous stamens $(3+3)$, Tricarpellary syncarpous, Axile placentation. $\oplus \mathbin{\rlap{\bigcirc}\kern-0.1em\raisebox{0.15ex}{\small\textbf{+}}} \overbrace{P_{(3+3)} A_{3+3}} \underline{G}_{(3)}$	• **Ornamentals:** Tulip, Gloriosa • **Medicine:** Aloe • **Vegetables:** Asparagus • **Colchicine:** *Colchicum autumnale* • **Food/Bulbs:** Onion, Garlic
Brassicaceae (Mustard Family)	Tetradynamous stamens $(2+4)$, Parietal placentation with false septum (Replum), Cruciform corolla. $\oplus \mathbin{\rlap{\bigcirc}\kern-0.1em\raisebox{0.15ex}{\small\textbf{+}}} K_{2+2} C_{4} A_{2+4} \underline{G}_{(2)}$	Mustard, Radish, Turnip, Cabbage, Cauliflower

7. HIGH-YIELD PRACTICE SHEET (NEET SELECTED)

Q1. Free central placentation and polyadelphous / diadelphous stamens are seen respectively in:

(A) Dianthus and Pea
(B) China Rose and Tomato
(C) Mustard and Argemone
(D) Sunflower and Marigold

Q2. Pneumatophores, Stilt roots, and Prop roots are found respectively in:

(A) Banyan, Sugarcane, Rhizophora
(B) Rhizophora, Sugarcane, Banyan
(C) Sweet Potato, Turnip, Maize
(D) Rhizophora, Banyan, Sugarcane

Q3. Match the plant modification with its exact NCERT example:

(A) Phylloclade → Opuntia
(B) Phyllode → Australian Acacia
(C) Offset → Eichhornia
(D) All of the above are correct

ANSWER KEY & EXPLANATIONS:
- **Q1 → (A) Dianthus has Free Central placentation; Pea has Diadelphous stamens.**
- **Q2 → (B) Pneumatophores = Rhizophora, Stilt roots = Sugarcane/Maize, Prop roots = Banyan Tree.**
- **Q3 → (D) All pairs are 100% correct NCERT direct matches.**